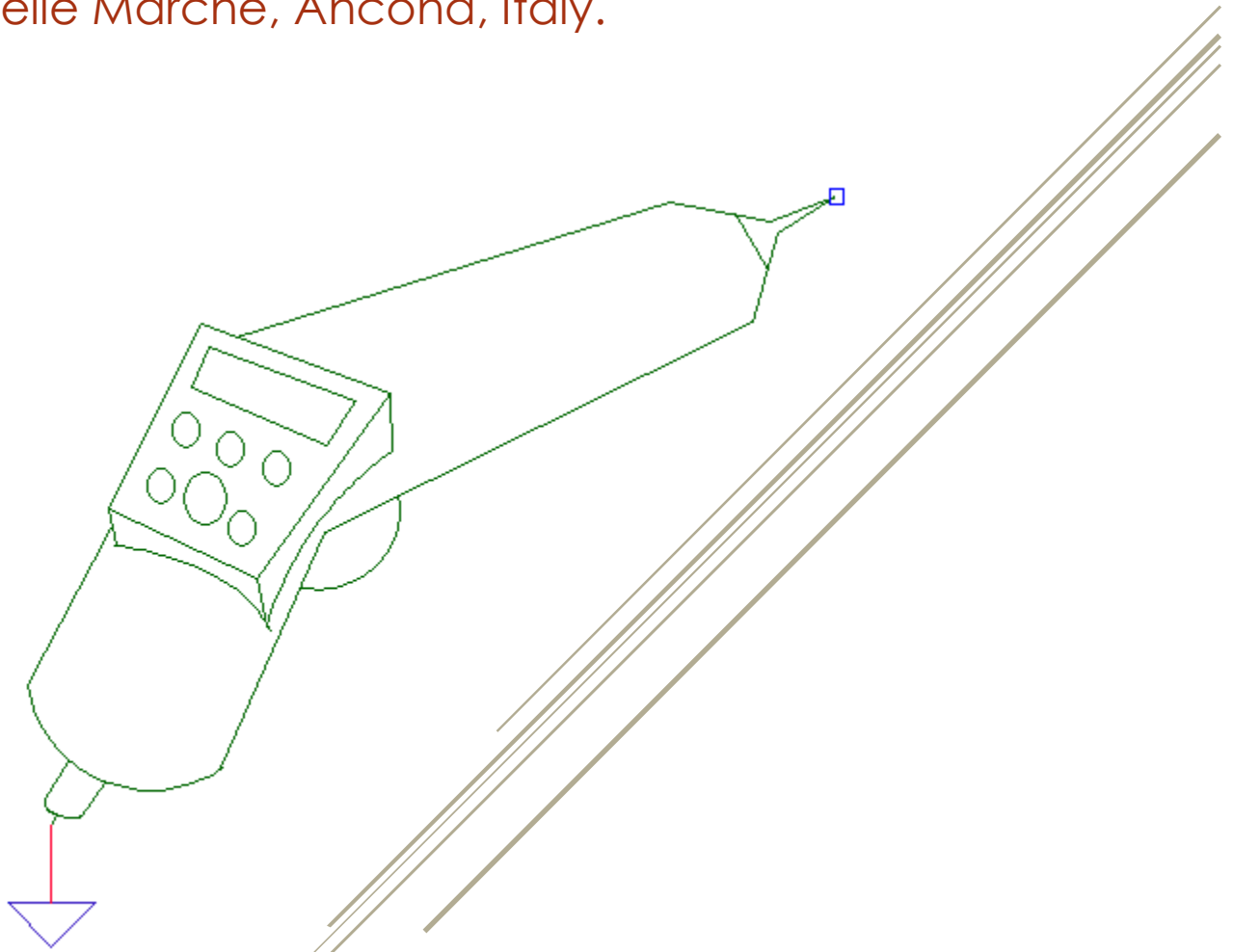


A SIMPLE SPICE ESD GENERATOR CIRCUIT BASED ON IEC61000-4-2 STANDARD

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ABSTRACT

The Electrostatic Discharge phenomenon is a great threat to all electronic devices and ICs. An electric charge passing rapidly from a charged body to another can seriously harm the last one.

The commonly accepted Electrostatic Discharge current waveform is the one set by the IEC 61000-4-2 [1].

This brief article describe a simple circuit, based on IEC 61000-4-2 [2] that define the typical waveform of the discharge current besides the range of test level, to estimate the possible effect of ESD at design time. This can help engineer to evaluate possible solutions during circuit (SPICE) simulation before perform expensive ESD tests.

WHAT IS ESD?

Static charge is an unbalanced electrical charge at rest. Typically, it is created by insulator surfaces rubbing together or pulling apart. One surface gains electrons, while the other surface loses electrons. This results in an unbalanced electrical condition known as static charge.

When a static charge moves from one surface to another, it becomes ESD. ESD is a miniature lightning bolt of charge that moves between two surfaces that have different potentials. It can occur only when the voltage differential between the two surfaces is sufficiently high to break down the dielectric strength of the medium separating the two surfaces.

When a static charge moves, it becomes a current that damages or destroys gate oxide, metallization, and junctions. ESD can occur in any one of four different ways: a charged body can touch an IC, a charged IC can touch a grounded surface, a charged machine can touch an IC, or an electrostatic field can induce a voltage across a dielectric sufficient to break it down [3].

ESD STRESS MODELS

ESD can have serious detrimental effects on all semiconductor ICs and the system that contains them. Standards are developed to enhance the quality and reliability of ICs by ensuring all devices employed have undergone proper ESD design and testing, thereby, minimizing the detrimental effects of ESD. Three major stress methods are widely used in the industry today to describe uniform methods for establishing ESD withstand thresholds (highest passing level).

HUMAN BODY MODEL (HBM)

The HBM is a component level stress developed to simulate the action of a human body discharging accumulated static charge through a device to ground, and employs a series RC network consisting of a 100 pF capacitor and a 1500 Ω resistor.

CHARGED DEVICE MODEL (CDM)

The CDM is a component level stress that simulates charging and discharging events that occur in production equipment and processes. Potential for CDM ESD events occur when there is metal-to-metal contact in manufacturing.

SYSTEM LEVEL ESD (IEC 61000-4-2)

The IEC system level ESD is a widely accepted European standard that defines an ESD event that is meant to be tested on actual end equipment to simulate a charged person or object discharging into electronic systems. The IEC standard defines an ESD stress that is much stronger than the component level ESD stresses defined by HBM and CDM.

The engineer must design following IEC 61000-4-2 standard to be able to declare the conformity CE ("Conformité Européenne").

The ESD generator circuit realized in this work is compliant with ESD generator used in EMC laboratories to perform CE-conformity tests, thus a SPICE simulation can be used to test ESD-immunity solutions before circuit production.

IEC 61000-4-2 WAVEFORM CHARACTERISTICS

The standard accurately describe the characteristics and performances of the ESD generator as well as the current waveform parameters.

The ESD phenomenon is a very short but very strong current transient and is represented in Figure 1.

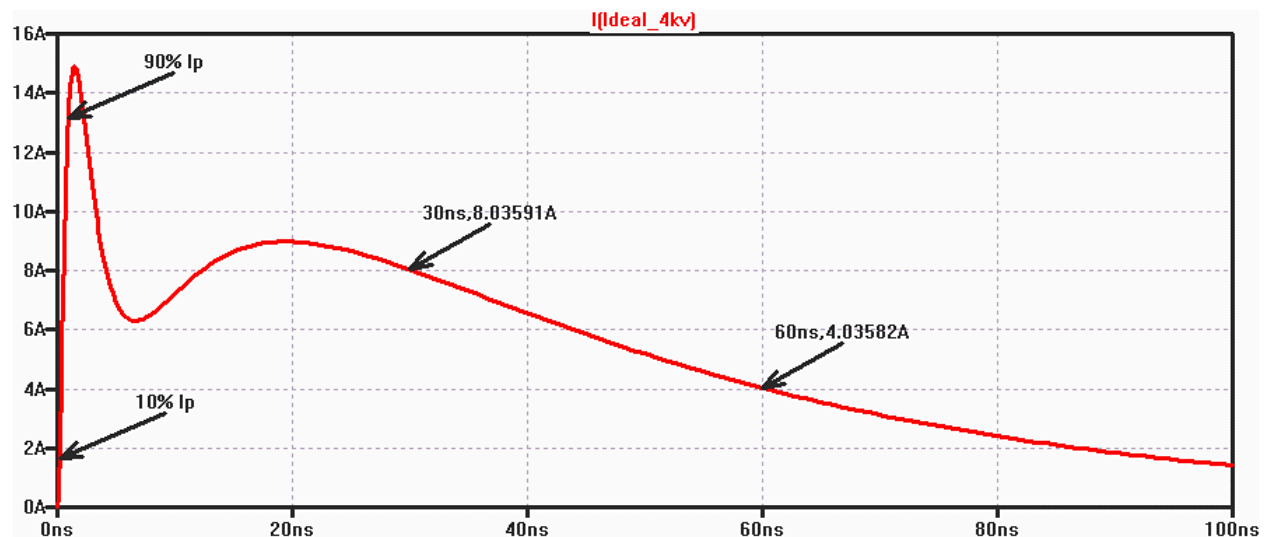


Figure 1: 61000-4-2 ideal contact ESD waveform at 4kV

This pulse is divided into two parts: The first peak, known as the "Initial Peak", is caused by the discharge of the arm, and generates the maximum current. The second peak is caused by the discharge of the body. The rise time of the initial peak is between 0.6 ns and 1 ns, and its amplitude depends on the charging voltage of the ESD simulator [1].

The standard describe a formula and the mains current level of the waveform for different test level.

Level	Indicated voltage	First peak current of discharge $\pm 15\%$	Rise time t_r ($\pm 25\%$)	Current ($\pm 30\%$) at 30 ns	Current ($\pm 30\%$) at 60 ns
1	2 kV	7.5 A	0.8 ns	4 A	2 A
2	4 kV	15 A	0.8 ns	8 A	4 A
3	6 kV	22.5 A	0.8 ns	12 A	6 A
4	8 kV	30 A	0.8 ns	16 A	8 A

The reference point for measuring the time for the current at 30 ns and 60 ns is the instant when the current first reaches 10 % of the 1st peak of the discharge current.

NOTE The rise time, t_r , is the time interval between 10 % and 90 % value of 1st peak current.

Table 1: contact discharge current waveform parameters

Even though the IEC 61000-4-2 [2] include a simplified circuit of ESD generator, it is incompatible with the discharge current equation described in the standard and the waveform shown in Figure 1. A PSpice software simulation can prove this [4].

THE BASIC ESD GENERATOR MODEL

In order to run ESD stress simulation, an ESD-generator model was built. This can help engineers to test different solutions in a SPICE simulator to overcome strength over-voltages before realizing PCB circuit and test it against ESD.

The objective is to generate an ESD pulse that accurately corresponds to the current stress waveforms at various stress levels in accordance with the IEC 61000-4-2 specification.

The ESD can be simulated by two parallel R, L, C circuits with charged capacitors. The generator equivalent circuit is shown in Figure 2.

Here the standardized network elements of the ESD-generator are represented by R1 (330ohm) and C1 (150pF). The inductor L1 is considered to be the obligatory ground strap with the length of about 2 m. Physically the first peak of the pulse is shaped by additional lumped and parasitic elements around and in the tip of the ESD-generator [5].

Note that the values of R, L, and C for both branches are tweaked to correctly represent standard IEC stress waveform; ".ic V(c1)=4kV V(c2)=4kV" refers to the initial condition of the voltage on the capacitors for a 4kV zap.

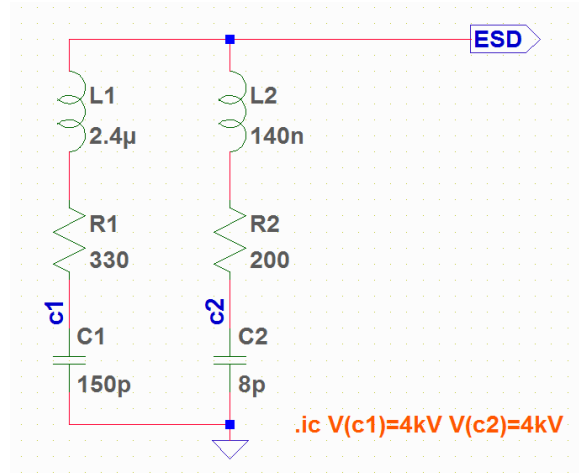


Figure 2: The general equivalent circuit of basic IEC61000-4-2 generator model

According to the IEC 61000-4-2 standard, the current waveform has to be verified with a low ohmic current sensor. In fact, about the measurement of discharge current, the standard write: "coaxial current target used to measure the discharge current of ESD generators shall have an input impedance, measured between the inner electrode and ground, of no more than 2.1 Ω at d.c." [2]. Thus, in order to measure the simulated discharge current, a 2-ohm resistor is used.

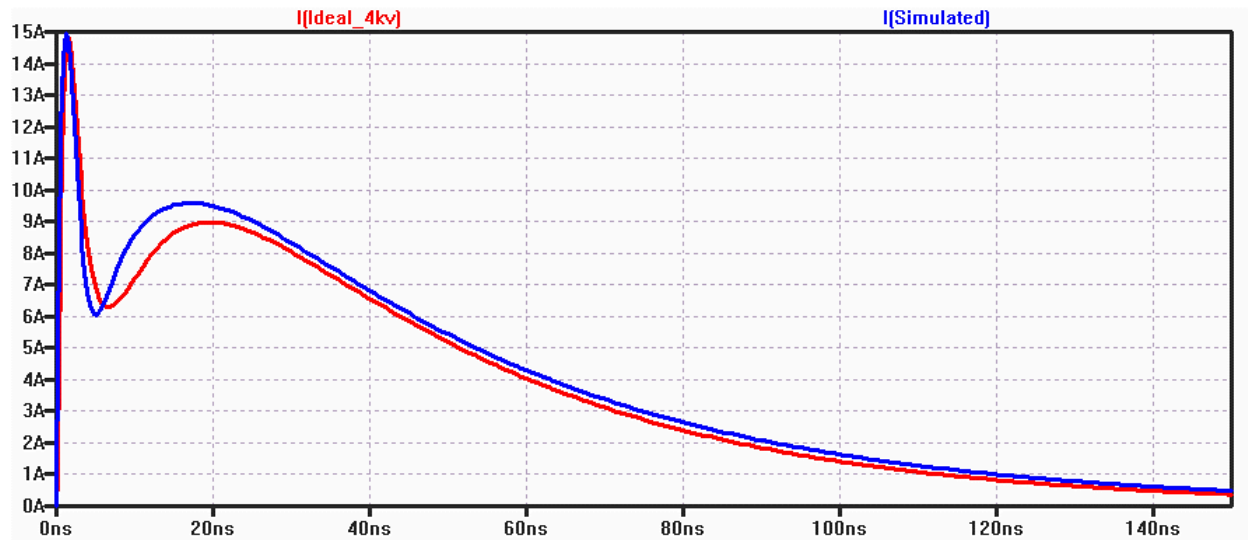


Figure 3: The ideal ESD waveform and the SPICE-simulated ESD at 4kV level

The SPICE simulated ESD waveform is shown in Figure 3 in blue; the red waveform is the ideal ESD waveform, as described in the specification's formula. As can be seen, there is a good matching between the two waveform, in fact, the simulated currents at the first peak, 30ns and 60ns meet the IEC 61000-4-2 specification.

The Table 2 describe the waveform characteristics at different test levels (2 kV, 4kV, 6kV and 8kV) of the simulated generator versus ideal current waveform.

Level	Indicated voltage	First peak current of discharge $\pm 15\%$	Rise time t_r ($\pm 25\%$)	Current ($\pm 30\%$) at 30 ns	Current ($\pm 30\%$) at 60 ns
1 ideal	2 kV	7.5 A	0.8 ns	4 A	2 A
1 simulated	2 kV	7.48 A	0.75 ns	4.17 A	2.14 A
2 ideal	4 kV	15 A	0.8 ns	8 A	4 A
2 simulated	4 kV	14.97 A	0.75 ns	8.33 A	4.29 A
3 ideal	6 kV	22.5 A	0.8 ns	12 A	6 A
3 simulated	6 kV	22.46 A	0.75 ns	12.5 A	6.43 A
4 ideal	8 kV	30 A	0.8 ns	16 A	8 A
4 simulated	8 kV	29.95 A	0.75 ns	16.65 A	8.58 A

Table 2: The waveform characteristics at different test levels

THE (LT)SPICE MODEL

The circuit was build using LT-Spice, a freeware SPICE simulator of electronic circuits but could be redesigned using other SPICE simulators.

The ESD-generator proposed is a parametric circuit, thus could be chosen:

- the IEC61000-4-2 test level (the ESD voltage) by setting the parameter "IEC61000_4_2_test_level" from 1 to 4 (see Table 2).
- the starting time of the ESD simulated phenomena by setting the parameter "td" with time in seconds.

The circuit shown in Figure 4 is the same as in Figure 2 with the addition of two voltage-controlled switch that implements the control on ESD starting time.

Control on ESD voltage is done setting the initial voltage condition of the two capacitor (see string ".ic V(c1)=4kV V(c2)=4kV").

The circuit is saved on file: "IEC61000-4-2_ESD_Generator.asc"

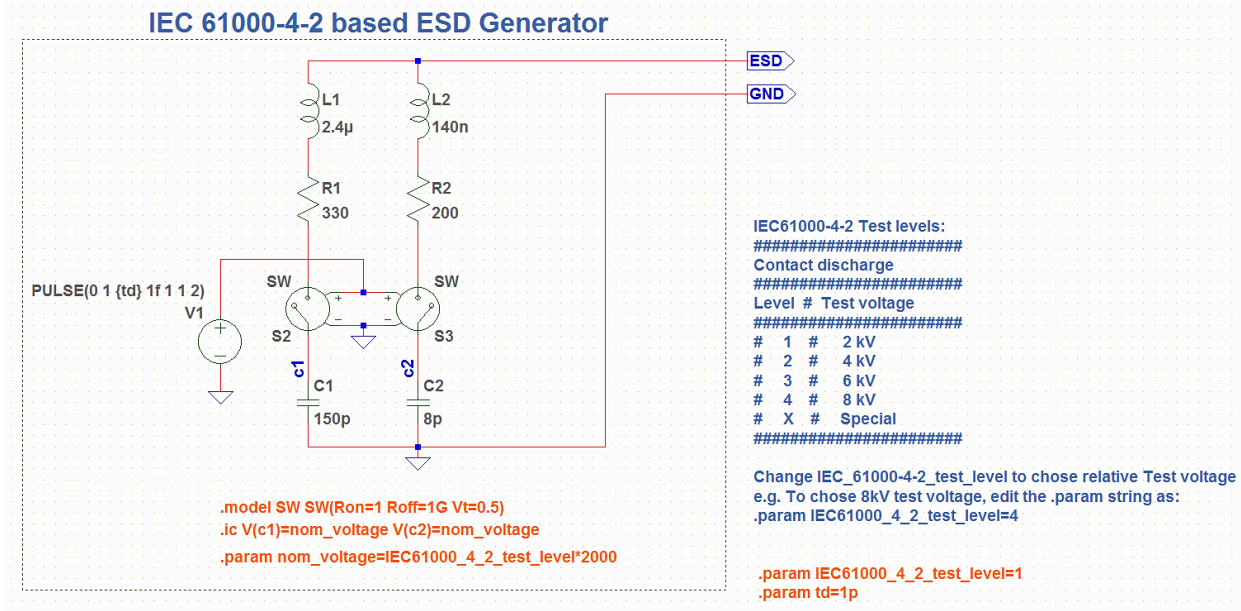


Figure 4: the ESD generator SPICE circuit

To simplify the use of the circuit, a symbol for this generator similar to ESD gun used in EMC laboratories is designed. It is available on file "IEC61000-4-2_ESD_Generator.asy". To use the symbol, create a new folder and copy IEC61000-4-2_ESD_Generator.asy and IEC61000-4-2_ESD_Generator.asc files inside. Then, create a new LT-Spice schematic and add a new component selecting the top directory containing the schematic (see Figure 5). This folder will also contain the ESD generator symbol.

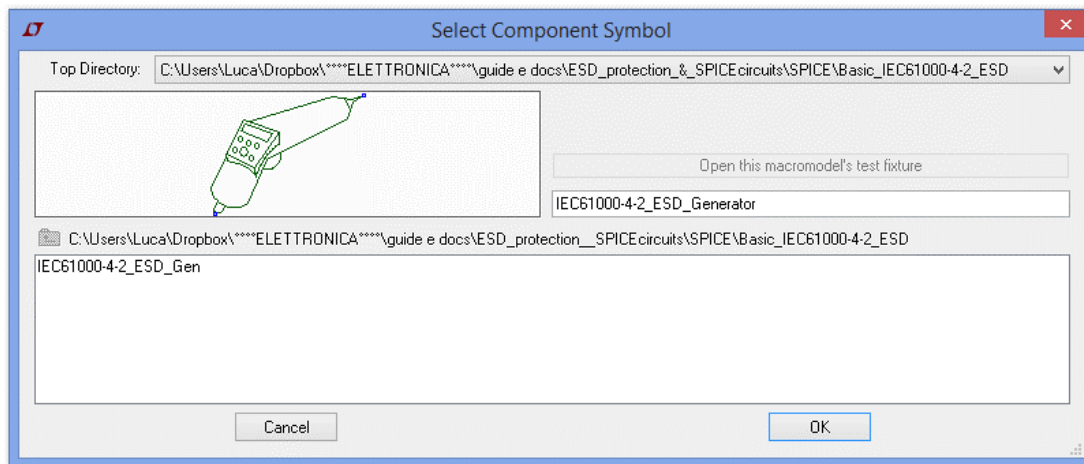


Figure 5: the select component symbol window

To set the parameters just right-click on the ESD-gun symbol and copy the string "IEC61000_4_2_test_level=<desired_level> td=<desired_time_delay>" on PARAMS text-box. Obviously change the <desired_level> and <desired_time_delay> with desired value as described in the schematic (see the blu text near the ESD-gun in Figure 6).

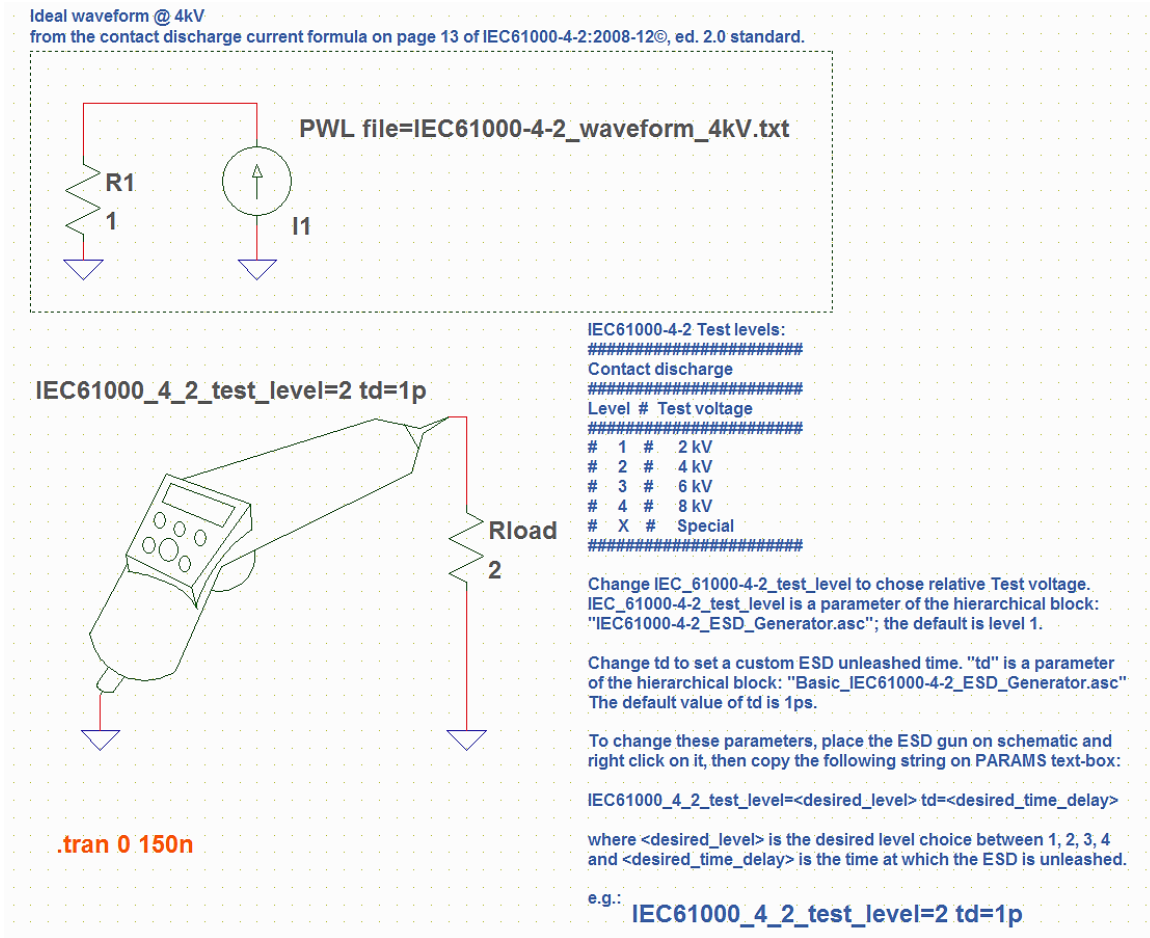


Figure 6: A schematic example

As visible in Figure 6, in "Example_IEC61000-4-2_ESD_circuit.asc", there are the simulated ESD-generator and I1, a behavioral-current-generator. This current generator read the data from "IEC61000-4-2_waveform_4kV.txt" that is a PWL file containing data generated with Matlab[®] using IEC61000-4-2 ESD current formula.

Plotting $I(R1)$ and $I(Rload)$, it's easy to compare the simulated current and the IEC61000-4-2 current (referred to level 2, 4kV ESD only) as shown in Figure 3.

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